精華摘要（擴充版）

A. Attention Getter（驅動與動機）
電商售後問題包含換貨、退貨、退款、維修與運費詢問，占淘寶客服聊天的 56.5%，是客服最大宗的需求。這項數字來自阿里巴巴的現場實驗，涵蓋 5,940 名客服與約 256 萬則對話（Ni et al., arXiv）。台灣 2025 年零售網路銷售額達 6,716 億元，占零售業約 13.9%（工商時報）。美國線上銷售的退貨率約 19.3%（Richpanel 引述 NRF）。需求大、重複性高，因此成為 AI 導入的首選場景，但也最容易出錯。[1]

B. But（挑戰與現有限制）

- **P1 幻覺與拒答失效：** 評測 20 個前沿 LLM 發現，放大模型規模幾乎無助於拒答，推理微調甚至使拒答平均下降 24%（AbstentionBench）。Cursor 的客服機器人曾捏造不存在的政策，引發用戶取消訂閱（Wired）。[2]
- **P2 法律責任：** Air Canada 的聊天機器人誤導乘客，法庭判企業敗訴並駁回「機器人是獨立實體」的辯解（Law360 Canada）。台灣消保法規定七日解除權與十五日內返還價金（全國法規資料庫）。[3]
- **P3 權限與個資：** PCI SSC 指出接觸帳戶資料的 AI 屬於 PCI DSS 範圍，處理明文卡號需人工核准（Jazz Security 整理）。OWASP 也將「過度代理」列為主要風險（OWASP）。[4] [5]
- **P4 顧客體驗：** 87% 的顧客要求保留真人管道（Gartner）。有明確真人選項時，55% 的人信任 AI，沒有時只剩 26%（Five9）。

C. Cure（解決方案）
本研究提出 ShopAssist，一個依據知識庫回答購物問題的客服 Domain AI Agent。它具備四項精準處置能力：回答必附來源、證據不足拒答、缺少資訊追問，以及敏感與越權請求建立模擬工單。設計重點是讓 AI 在能力範圍內回答，範圍外則拒答、追問或轉交，而不是追求全自動。

D. Development（系統發展與整合）
系統整合五個層次：

- 意圖路由： 13 條規則，先判斷問題類型與風險。
- 個資遮蔽： 自動遮蔽敏感資訊，避免進入模型。
- 混合檢索： BM25 加稠密檢索，以 RRF 融合。混合檢索在金融問答基準上優於單一方法（arXiv 2604.01733）。[6]
- 雙重證據閘門： 證據不足時直接拒答，而不是交給模型猜測。
- 生成後依據檢查： 回答產生後再核對是否有證據支持。

這樣設計的理由是，拒答能力不能只靠提示詞，需要系統層面的機制。

E. Experiments（實驗設計）
建立 78 題基準資料集，涵蓋 easy、medium、hard 三種難度與 8 大類別。比較三種變體：

A_llm_only（純模型）、B_rag_basic（基本檢索增強）與 C_shopassist（完整系統）。另進行 6 組消融實驗，逐一移除元件，找出各機制的實際貢獻。這一點很重要，因為混合檢索的效果並不一致，在某些任務中單純融合反而輸給 BM25（arXiv 2609.34776）。[7]

F. Findings（預期成果）
Proposal 階段設定三項關鍵假設 H1–H3，預期證明 ShopAssist 能顯著降低不安全回答率、提升拒答召回率，並量化其代價，例如過度拒答、延遲與轉人工比率。代價需要被量化，因為阿里巴巴的實驗顯示，代理式 AI 縮短處理時間的同時，AI 可處理對話的客戶評分下降 0.858 分（arXiv 2605.14830）。預期成果能為企業提供安全、合規且可轉人工的客服部署依據，也補上中文與台灣法規情境下的評測資源。

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1. https://www.ctee.com.tw/search/電商產業
2. https://www.wired.com/story/cursor-ai-hallucination-policy-customer-service/
3. https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=J0170001
4. https://owasp.github.io/www-project-top-10-for-large-language-model-applications/assets/PDF/OWASP-Top-10-for-LLMs-v2025.pdf
5. https://jazz.security/dlp-explained/dlp-basics/can-support-agents-use-ai-with-customer-data/
6. https://arxiv.org/html/2604.01733
7. https://arxiv.org/html/2609.34776v1
8. https://arxiv.org/pdf/2508.18760v3.pdf
9. https://www.thedailystar.net/tech-startup/news/ai-first-company-now-rehiring-human-staff-after-quality-drop-3892896
10. https://www.mirai.com.tw/category/diagram/reports-e-commerce/
11. https://mintnl.substack.com/p/ai-can-do-all-the-work-but-whos-responsible
12. https://arxiv.org/html/2608.03860
13. https://trec.nist.gov/pubs/trec34/papers/cfdalab.rag.pdf
14. https://aclanthology.org/2026.bionlp-1.10/
15. https://arxiv.org/html/2607.24663
16. https://jutif.if.unsoed.ac.id/index.php/jurnal/article/view/5755
17. https://www.iea.nl/sites/default/files/2026-09/R&D_Report_Information Retrieval Using Retrieval-Augmented Generation_0.pdf
18. https://www.ibtimes.co.uk/ai-boomerang-companies-rehire-workers-after-automation-1829393
19. https://jutif.if.unsoed.ac.id/index.php/jurnal/article/download/5755/1425/25957
20. https://ar5iv.labs.arxiv.org/html/2409.01357
21. https://law.moj.gov.tw/LawClass/LawSingle.aspx?flno=19&pcode=J0170001
22. https://law.moj.gov.tw/LawClass/LawSingleRela.aspx?FLNO=19&PCODE=J0170001&ty=L
23. https://law.moj.gov.tw/LawClass/LawParaDeatil.aspx?media=print&pcode=J0170001&bp=2
24. https://law.moj.gov.tw/LawClass/LawParaDeatil.aspx?bp=5&pcode=J0170001
25. https://www.ey.gov.tw/NewOpenData/XML/193

26. https://www.sme.gov.tw/consumer/files/10623/83DEB66F-E070-4727-92BD-5558EB615957
27. https://www.ey.gov.tw/Page/24C4B877E850ED4E/0f556196-c5e1-4113-895e-c5c1ae9c2c84
28. https://www.entrepreneur.com/business-news/klarna-ceo-reverses-course-by-hiring-more-humans-not-ai/491396
29. https://cpc.ey.gov.tw/Page/4432D6D5FA6677B9/2cd52250-9cc5-4c34-99ee-a350985d0b37
30. https://www.sme.gov.tw/consumer/files/6104/E05146A8-D67E-4D5E-A739-A6150AFC4BDA
31. https://www.hcc.gov.tw/upload/web_program/2026/04/13/個人資料保護法1141111_151807.pdf
32. https://www.tpipas.org.tw/resource_download.aspx?tag=D&tno=2482
33. https://www.pdpc.gov.tw/News_Html/100/
34. https://www.pdpc.gov.tw/
35. https://join.gov.tw/policies/detail/7c247cf0-664d-4402-8a27-fd620119deea
36. https://uract.nlma.gov.tw/rule/detail/601
37. https://www.nhrm.gov.tw/WebUPD/nhrm/News/「個人資料保護法部分條文」修正草案總說明及條文對照表.pdf
38. https://gazette.nat.gov.tw/EG_FileManager/eguploadpub/eg032015/ch01/type3/gov79/num3/Eg.pdf
39. https://terencekok.com/research/case-studies/klarna-ai-assistant/pdf/
40. https://www.pdpc.gov.tw/News_Content/101/429/
41. https://www.leeandli.com/TW/NewslettersDetail/7584.htm
42. https://airisk.mit.edu/ai-incidents/1039-anysphere-ai-support-bot-for-cursor-reportedly-invents
43. https://www.sans.org/newsletters/newsbites/xxvii-31
44. https://fortune.com/article/customer-support-ai-cursor-went-rogue/
45. https://www.useheadlights.com/library/HD-INC-005.md
46. https://news.ycombinator.com/item?id=43683012
47. https://externalcontrollayer.ai/record-18
48. https://github.com/vectara/awesome-agent-failures/blob/main/docs/case-studies/cursor-sam-support-bot.md
49. https://www.tribtoday.com/news/business/2026/11/what-to-consider-if-a-company-that-made-ai-related-job-cuts-wants-you-back/
50. https://cursor.com/terms-of-service
51. https://www.theregister.com/special-features/2025/04/18/cursor-ai-support-bot-hallucinated-its-own-company-policy/1015579
52. https://personalagentbench.com/benchmarks/tau-bench/
53. https://www.honeyhive.ai/learn/tau-bench
54. https://github.com/JevAgentHarness/JevAgentHarness/blob/main/research/docs/TAU2.md
55. https://github.com/PatrickWiloak/ai-zero-to-hero/blob/main/explainers/benchmarks/agents-and-computer-use.md
56. https://personalagentbench.com/benchmarks/tau-bench.md
57. https://arxiv.org/html/2506.07982
58. http://agentbeats.dev/agentbeater/tau2-bench?leaderboard_page_0=7
59. https://arxiv.org/html/2406.12045
60. https://fortune.com/2025/05/09/klarna-ai-humans-return-on-investment/

61. https://github.com/topics/tau2-bench
62. https://github.com/sierra-research/tau2-bench
63. https://genai.owasp.org/initiatives/top-10-for-llm-and-genai/
64. https://owasp.org/projects/top-10-for-large-language-model-applications?tab=example
65. https://genai.owasp.org/llm-top-10/
66. https://owasp.github.io/DevSecOpsGuideline/3-Governance/3-4-AI-Governance-and-Risk/
67. https://genai.owasp.org/llm-top-10-2023-24/
68. https://genai.owasp.org/llmrisk2023-24/llm08-excessive-agency/?ref=blog.passwork.club
69. https://genai.owasp.org/llm-top-10/page/5/
70. https://www.cxnetwork.com/artificial-intelligence/articles/the-cost-of-cost-reduction-an-ai-governance-playbook-for-customer-leaders
71. https://aws.amazon.com/blogs/machine-learning/secure-a-generative-ai-assistant-with-owasp-top-10-mitigation/
72. https://genai.owasp.org/2026/10/08/genai-and-agentic-ai-exploit-roundup-q3-2026/
73. https://www.securisea.com/resources/pci-compliance-and-ai-managing-new-compliance-risks
74. https://theworldofai.org/ai-compliance/pci-dss-ai/
75. https://www.cognigy.com/product-updates/pci-dss-compliance
76. https://jazz.security/dlp-explained/dlp-basics/pci-dss-compliance-checklist/
77. https://www.bland.ai/compare-markdown/landing-pci.md
78. https://www.reuters.com/practical-law-the-journal/transactional/pci-dss-compliance-2026-05-01/
79. https://github.com/GenAI-Security-Project/crosswalk/blob/main/agentic-top10/Agentic_PCIDSS.md
80. https://economictimes.indiatimes.com/tech/technology/klarna-retreats-from-ai-only-customer-service-push-resumes-hiring-amid-quality-concerns-report/comment/121267347.cms
81. https://services.google.com/fh/files/misc/gcp_pci_dss_v4_responsibility_matrix.pdf
82. https://telnyx.com/voice-ai-agents/pci-compliant-payments
83. https://www.hostinger.com/ph/tutorials/ai-agent-for-business/
84. https://dxc.com/insights/knowledge-base/paper/agentic-ai-workforce-transformation
85. https://www.gartner.com/en/newsroom/topics/customer-service-support
86. https://www.cmswire.com/customer-experience/just-4-of-customer-service-leaders-hit-their-expected-automation-returns/
87. https://finance.yahoo.com/technology/ai/articles/surveil-recognized-pioneer-october-2026-212500980.html
88. https://www.gspann.com/insights/blog/the-agentic-ai-governance-mistake-move-fast-and-pay-thrice
89. https://www.gartner.com/en/newsroom/press-releases/2026-09-29-gartner-predicts-70-percent-of-enterprises-will-abandon-agentic-ai-built-by-vendor-forward-deployed-engineering-by-2028
90. https://aibusinessweekly.net/p/ai-agent-platforms
91. https://www.fintechweekly.com/news/klarna-hires-customer-service-after-ai-pivot
92. https://www.gartner.com/en/newsroom/press-releases/2026-11-05-gartner-survey-finds-21-percent-of-service-and-support-leaders-have-reduced-human-agent-headcount-because-of-ai
93. https://money.udn.com/money/story/5635/9290776

94. https://blog.104.com.tw/taiwan-retail-ecommerce-market-2026/
95. https://branch8.com/zh-TW/posts/taiwan-ecommerce-2026-market-data-report
96. https://www.mirai.com.tw/taiwan-e-commerce-key-market-data-and-forecasts/
97. https://www.mirai.com.tw/reports/
98. https://www.mirai.com.tw/category/diagram/reports-logistics/
99. https://www.mirai.com.tw/2026-annual-overview-of-taiwans-retail-and-e-commerce-sub-sector-structure-diagram/
100. https://www.deloitte.com/tw/tc/Industries/consumer/perspectives/taiwan-retail-next-growth-wave.html